Written exam, first call BIS

14th of April, 2025

Name _____

Exercise 1 (2.5 points)

Consider the following MA(1) random process:

$$Y_t = \epsilon_t + \theta\epsilon_{t-1}$$

where, ϵ_t is Gaussian white noise of mean 0 and variance $(0{,}1)^2$, and $\theta = 0{,}1$.

1. What is the expectation of the process $E(Y_t)$?
2. Calculate the variance $Var(Y_t)$.
3. Calculate the covariance $Cov(Y_t, Y_{t-1})$.
4. Calculate the covariance $Cov(Y_t, Y_{t-2})$.
5. [**R**] Simulate an observation of size 1000 of this MA(1) process and visualize it.

Exercise 2 (3 points)

Object *series* is a time series of length 1000. In order to load object *series*, download file *series.RData* from eGela and execute *load("series.RData")*. Set a fixed random seed (*set.seed(300)*) before starting the exercise and execute all the lines consecutively.

1. [**R**] By observing the plot of the time series would you say it is stationary? Why?
2. [**R**] Check the previous result using a suitable statistical test for stationarity? What is the conclusion?
3. [**R**] Use the ACF plot to decide the order of the MA component of the series and justify your choice.
4. [**R**] Fit a MA(2) model to the time series. How many parameters does the model have? What are their values? What is the value of the AIC?
5. [**R**] Obtain one-step ahead predictions for a prediction horizon of 10 using the model in section 4 and visualize the time series and the forecasts.
6. [**R**] Fit a random forest model with a sliding window of size 5 to the *series* time series and obtain one-step ahead predictions for the next 10 time steps. Visualize the results together with the forecasts obtained by the MA(2) model.

Exercise 3 (3 points)

Object *tsdatabase* is a database of 12 time series of size 4. In order to load object *tsdatabase*, download file *tsdatabase.RData* from eGela and execute load("tsdatabase.RData").

1. Calculate the DTW distance between the first two time series of the dataset.
2. [**R**] Calculate this same distance using package TSdist. Do you obtain the same value?
3. [**R**] Perform clustering using the DTW distance and the KMedoids algorithm with k=3.
4. [**R**] Perform clustering using the DTW distance and the hierarchical clustering algorithm. Draw the dendogram and obtain the clustering result with k=3.
5. [**R**] Perform feature based clustering by using the *tsfeatures* package and the hierarchical clustering method with k=3. Before performing the clustering remove any features with NA values.
6. [**R**] If we assume that the ground truth clustering is the following:

   ```
   ground_truth <- c(2, 3, 1, 1, 2, 3, 3, 2, 3, 2, 1, 1)
   ```

   Evaluate the three clustering solutions and decide which is the best.

Exercise 4 (1.5 points)

Describe the differences between feature based, model based and distance based classification of time series and the advantages and disadvantages of each one.